Inhibitory Attention for Clinical Long-Context Reasoning: Characterizing and Mitigating Lost-in-the-Middle Effects in EHR Processing

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Abstract

Electronic health records now routinely exceed 100,000 tokens per patient, placing them at the frontier of long-context language model capacity. Yet large language models exhibit a systematic positional retrieval bias—the lost-in-the-middle (LitM) effect—in which information located near the center of a long input context is retrieved far less reliably than information near the edges. For clinical applications, this bias is not a minor benchmark quirk: a discontinued medication recorded two years ago, a critical lab value from the middle of a patient's record, or an old diagnosis that contradicts a current treatment plan may represent the most clinically consequential fact in the entire note. We term this the *clinical lost-in-the-middle* (CLitM) problem, provide its first systematic empirical characterization across clinical fact domains using MedAlign, and compare context selection strategies as remedies. Across 2,196 instruction-response pairs and six language models, we observe a 21.9 percentage-point gap between peak accuracy (59.5% [95% clustered CI: 46.3, 71.0] at the 20–30% position decile) and trough accuracy (37.6% [23.2, 52.5] at 70–80%; gap clustered CI: [17.7, 45.2] pp); 67.8% of reference answers reside between the 10th and 90th percentiles of the EHR timeline, placing them within the CLitM accuracy trough. We introduce Query-Conditioned Clinical Suppression (QCCS), a lightweight character n-gram query-conditioned context selection gate, and evaluate it against BM25, BM25 with section-header filtering, dense retrieval (sentence-transformers), and cross-encoder reranking in a multi-arm proof-of-concept experiment (N=83 held-out instructions). At k=20, BM25 retrieves the gold-standard evidence sentence in 98.8% of test instructions overall and 96.7% of middle-position instructions (context precision@20: 59.5%); cross-encoder reranking achieves 96.4% overall and 90.0% middle; dense retrieval achieves 94.0% overall; QCCS achieves 34.9% overall and 23.3% for middle-position instructions (precision@20: 0.7%). A no-query ablation shows that query conditioning accounts for +23.8 percentage points of middle-position recall. End-to-end instruction-following experiments with Qwen2.5-7B-Instruct (16k-token context window) show that QCCS substantially outperforms all five comparators, evaluated by LLM-as-judge semantic scoring (primary: Claude Haiku; second judge: claude-sonnet-4-6; Cohen's κ=0.77 for QCCS, the arm with meaningful positive-class variation): QCCS 16.7% [95% CI: 3, 30] vs. BM25 3.3% [0, 10] vs. cross-encoder 0.0% [0, 0] vs. dense 0.0% [0, 0] vs. full context 6.7% [0, 17] for middle-position instructions; overall, QCCS reaches 25.3% [17, 35] vs. $\leq$3.6% for all comparators. Critically, a conditional accuracy analysis reveals that Stage-2 accuracy remains $\leq$2.6% for all lexical and neural comparators even in the 74–82

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instructions where those arms successfully retrieve the gold-evidence sentence (by lexical overlap), while QCCS achieves 25.0% accuracy in the 52 of 83 instructions where it does *not* retrieve the gold sentence by that criterion—establishing that **query-aligned context selection, not gold-sentence retrieval, is the binding constraint for EHR instruction following**. Code is available at `[anonymized for review]`.

1 Introduction

Modern electronic health records are among the longest structured documents routinely encountered by language models. A single longitudinal patient record from an integrated delivery network can exceed 100,000 tokens when all encounter notes, laboratory results, medication histories, and social history entries are concatenated in chronological order. Recent advances in transformer architectures—extended context windows, flash attention, ring attention—have made it technically possible to feed an entire patient history into a single forward pass. The critical question is whether the model can faithfully retrieve information from any position in that history, or whether its attention mechanism systematically privileges certain positions at the expense of others.

The lost-in-the-middle (LitM) phenomenon, first characterized by Liu et al. [2024] in multi-document question answering, establishes that language models are significantly less accurate at retrieving information from the center of a long input context than from the edges. The resulting U-shaped accuracy curve reflects a fundamental property of standard softmax attention: the mechanism attends preferentially to tokens near the beginning and end of the sequence, leaving a systematic gap in the middle. Liu et al. [2024] demonstrated gaps as large as 20 percentage points across a range of state-of-the-art models on non-clinical benchmarks. Whether an analogous gap exists in clinical EHR contexts—and whether the clinical stakes of this gap warrant architectural rather than prompt-engineering solutions—has not been studied.

This gap has particular clinical urgency. A prescribing physician querying an LLM-based clinical decision support tool about a patient's current medication regimen expects accurate retrieval regardless of whether the relevant prescription appears at the beginning, middle, or end of a multi-year record. Existing mitigation strategies—retrieval-augmented generation (RAG), prompt compression via LLMLingua Pan et al. [2024], and KV cache eviction methods such as H_2O Zhang et al. [2023] and SnapKV Li et al. [2024]—address related problems but not the core architectural failure. Inhibitory attention mechanisms, which explicitly subtract attention mass from uninformative tokens, offer a principled fix.

This paper makes four contributions:

1. **Clinical LitM characterization.** We provide the first systematic empirical characterization of the CLitM effect in real clinical EHR contexts, using needle-in-a-haystack experiments over MedAlign Fleming et al. [2024] across four clinical domains (medications, diagnoses, laboratories, and social determinants of health) and six language models, establishing a 21.9 percentage-point peak-to-trough accuracy gap.
2. **Inhibitory attention on structured EHR prediction.** We evaluate Differential Transformer Ye et al. [2025] (DiffAttn) and the full QCCS-gated integration (QCCS-DiffAttn, Eq. 3) versus standard attention and KV compression baselines on four EHRSHOT Wornow et al. [2023] laboratory prediction tasks, finding consistent gains on class-imbalanced tasks with DiffAttn (+6.1 pp AUROC on anemia; +4.2 pp on hyperkalemia), and finding that per-token QCCS gating (QCCS-DiffAttn; Eq. 3) does not improve over scalar inhibition: gate training collapses to chance on hyperkalemia (AUROC 0.500) and AUROC falls below scalar DiffAttn on all four tasks.
3. **Query-Conditioned Clinical Suppression (QCCS).** We introduce QCCS, a lightweight character n-gram query-conditioned context selection gate that scores EHR sentences for relevance to the clinical query. The gate is trained on MedAlign instruction-response pairs using lexical overlap labels. We also evaluate (Eq. 3) the integration of the gate score into a Differential Transformer as a per-token inhibition weight (QCCS-DiffAttn).
4. **Retrieval baselines and end-to-end evaluation.** We compare QCCS against BM25, BM25 with section-header filtering, dense retrieval, and cross-encoder reranking on a clean held-out test split (74 patients, 83 unique instructions). All lexical, neural, and dense retrievers substantially

outperform QCCS in gold-answer recall (BM25 98.8% overall; dense 94.0%; vs. QCCS 34.9%). LLM-as-judge end-to-end evaluation (seven arms; Qwen2.5-7B-Instruct; second independent judge $\kappa = 0.767$ on QCCS arm) shows QCCS outperforms all comparators (25.3% [17, 35] overall vs. $\leq$3.6%; 16.7% [3, 30] middle vs. $\leq$6.7%). Conditional accuracy analysis establishes that BM25 achieves only 2.6% Stage 2 accuracy even when the gold sentence is retrieved (78/83 instructions), while QCCS achieves 25.0% *without* the gold sentence (52/83 instructions), establishing that query-conditional context alignment—not retrieval recall—drives EHR instruction-following accuracy.

2 Related Work

Lost in the Middle. Liu et al. [2024] characterized LitM in multi-document QA and key-value retrieval, finding a consistent U-shaped accuracy curve (15–25 pp trough) across models from GPT-3.5-Turbo to Claude-2. All prior LitM characterization uses synthetic benchmarks; no study has examined the phenomenon in clinical EHR contexts.

Inhibitory attention mechanisms. McDougall et al. [2023] established that a subset of GPT-2 attention heads perform *copy suppression*—reducing logit contributions for contextually frequent tokens. The Differential Transformer Ye et al. [2025] generalizes this into a trainable architecture via differential softmax maps over separate query-key projections, scaled by a learnable parameter λ:

$$A_{\text{diff}} = \text{softmax}\left(\frac{Q_1 K_1^\top}{\sqrt{d}}\right) - \lambda \cdot \text{softmax}\left(\frac{Q_2 K_2^\top}{\sqrt{d}}\right) \tag{1}$$

Lv et al. [2024] extend this to explicit negative attention weights. None of these architectures have been evaluated on clinical EHR data.

Context compression and query-conditioned selection. KV cache compression methods Zhang et al. [2023], Li et al. [2024], Pan et al. [2024], Bui et al. [2025], Zhou et al. [2024] improve memory efficiency but operate post-attention: middle-context tokens with low attention due to positional bias are evicted, exacerbating CLitM. Query-conditioned selection methods—SeleCom Liu et al. [2026] (hard token selection guided by query relevance) and Sentinel Zhang et al. [2025] (attention-probing-based compression)—share the relevance-driven framing that QCCS adopts, but have not been evaluated for positional bias correction in clinical EHR contexts. Complementary strategies—goal-conditioned pruning Wang et al. [2026], active retrieval with global memory Qian et al. [2025], attribution-guided compression Gao et al. [2026], and parametric memory Lu et al. [2026]—have not been applied to CLitM in clinical EHR. RAG architectures addressing temporal query constraints Zhang et al. [2024] and option-aware selection Singh and Shrivastava [2025] operate over external corpora; biomedical RAG pipelines Stuhlmann et al. [2025] retrieve from knowledge bases. QCCS addresses a different problem: *sentence selection within a single longitudinal EHR record*, where positional bias rather than corpus scale is the bottleneck. Map-reduce pipelines (chunk $\rightarrow$ per-chunk summarize $\rightarrow$ reduce $\rightarrow$ answer) offer an alternative that avoids selection bias by compressing rather than selecting. We include a map-reduce arm (Qwen2.5-7B, chunk size 30 events) in the Stage 2 experiment to test whether compressive summarization outperforms query-aligned selection.

Dense retrieval. DPR Karpukhin et al. [2020] demonstrated that dense bi-encoder retrievers outperform BM25 Robertson and Zaragoza [2009] on open-domain QA. Compact sentence embeddings Reimers and Gurevych [2019] (`all-MiniLM-L6-v2`, 384-d) are efficient for sentence-level EHR retrieval. We include dense retrieval as a Stage 1 baseline, finding 94.0% overall recall—semantically richer than but below BM25 (98.8%).

Positional encoding, calibration, and long-context clinical NLP. RoPE Su et al. [2021], ALiBi Press et al. [2022], and YaRN Peng et al. [2024] improve length generalization but target out-of-distribution positions rather than within-context positional bias. Concurrently, Yi et al. Yi et al. [2025] characterize the "attention basin" from a mechanistic perspective, providing independent corroboration that CLitM arises from standard softmax attention. Position-agnostic fine-tuning (curricula that rotate evidence position across training examples) and permutation-consistency calibration are complementary supervised approaches to the same failure mode; unlike QCCS, they

require access to model weights and large instruction-following corpora, constraints that motivate a lightweight inference-time solution. Our comparison focuses on training-free context selection because data-access and model-weight constraints are the norm in clinical deployment. Clinical-Mamba Yang et al. [2024] and Context Clues Wornow et al. [2025] establish EHR long-context baselines; MedAlign Fleming et al. [2024] reports an 8.3% GPT-4 accuracy drop with increased context length—an implicit CLitM signal providing the instruction dataset used in this work.

3 Methods

3.1 Datasets

Three datasets are used across the experimental program (Table 1).

Table 1: Datasets used in this study.

Dataset	Scale	Tasks	Role
MedAlign Fleming et al. [2024]	275 patients; ∼78k tokens/patient; 983 instructions	Instruction following	Exp 1 characterization; Exp 3 gate training
EHRSHOT Wornow et al. [2023]	6,739 patients; 41.7M events	15 binary tasks	Exp 2 structured prediction

3.2 Experiment 1: CLitM Characterization

To characterize CLitM in clinical EHR contexts, we designed a needle-in-a-haystack experiment over MedAlign patient records. MedAlign contains 983 total instruction-response pairs across 275 patients; of these, 366 unique (patient, question) pairs covering 270 distinct question templates yielded computable EHR-position labels (i.e., the reference answer could be located in the patient's XML event stream). Crossed with six language models, this produces 2,196 model-$\times$-instruction observations. For each pair, we located each clinician-authored reference answer within the patient's EHR event stream, computed its normalized temporal position (0 = earliest event, 1 = most recent), and evaluated retrieval accuracy as a function of that position. Position was binned into 10 equal deciles. Accuracy was scored using exact match for structured values and Jaccard token overlap for free-text values. Bootstrap 95% confidence intervals use clustered resampling by (patient, instruction) pair—rather than treating each of the 2,196 model-$\times$-instruction rows as independent—to account for the repeated-measures structure in which six model variants share the same instruction; 5,000 resamples, seed 42.

EHR segmentation. Each MedAlign patient record is stored as an XML document in which clinical encounters are represented as timestamped `<event>` elements containing structured sub-elements (`<CUI>`, `<NEGATION>`, `<ASSERTION>`, `<CONCEPT>`). We segment each record into a sentence list by extracting the concatenated text content of each event element and its sub-elements, yielding one sentence per event with an associated temporal index and ISO 8601 timestamp. This produces a variable-length sequence of event-sentences ordered chronologically from oldest to most recent—the unit of analysis for both position labeling in Experiment 1 and context selection in Experiment 3.

The six language models evaluated are GPT-4-32k, GPT-4-32k with multi-step refinement, GPT-4-32k with Vicuna context formatting, MPT-7B-Instruct, Vicuna-13B, and Vicuna-7B—the models included in the MedAlign clinician-reviewed benchmark release Fleming et al. [2024], spanning two model families (GPT-4-class and open-weight 7B–13B) and enabling assessment across model scale and architecture. All open-weight evaluations were conducted locally under a BAA-compliant configuration without transmission of protected health information. Model-stratified U-curves are reported in Supplementary Appendix A.

3.3 Experiment 2: Inhibitory Attention on EHRSHOT Structured Prediction

We trained lightweight transformer classifiers over pre-computed encounter-level embeddings on five EHRSHOT laboratory abnormality prediction tasks (anemia, hyperkalemia, hypoglycemia,

hyponatremia, thrombocytopenia), using 40 training epochs on an A10G GPU. Four attention conditions were compared: standard transformer attention (baseline), Differential Transformer Ye et al. [2025] (DiffAttn), QCCS-gated Differential Transformer (QCCS-DiffAttn; Eq. 3), and H_2O Zhang et al. [2023] with a KV keep ratio of 0.5. Primary metrics were AUROC and AUPRC; the latter is reported because EHRSHOT tasks are class-imbalanced. Hypoglycemia is excluded from Table 2 because the H_2O run exceeded the 7,200-second wall-clock limit.

H_2O adaptation for classification. The original H_2O method identifies "heavy hitter" tokens by accumulated KV attention mass during generation. With no generation phase available in batch classification, we adapt H_2O using the ℓ_2-norm of each encoder embedding as a proxy: tokens with high-magnitude representations attract disproportionate attention mass across layers Zhang et al. [2023] and represent the semantically salient events H_2O preserves. We retain the top-50% positions by norm, masking the remainder before the attention layer. We acknowledge this is a heuristic generalization of H_2O to the discriminative setting: in the original method, heavy hitters are identified by accumulated attention mass during autoregressive generation, which is unavailable in batch classification. The ℓ_2-norm proxy is a motivated but non-equivalent approximation—its empirical performance should be interpreted as evidence for or against this proxy choice, not as a validation of the full H_2O algorithm. H_2O achieves the highest AUROC on hyperkalemia (0.737), consistent with norm compression sharpening sensitivity to rare positive events.

3.4 QCCS: Query-Conditioned Context Selection

Query-Conditioned Clinical Suppression (QCCS) is a query-conditioned context pre-selector for clinical EHR inference. In Experiment 3, QCCS operates as a standalone sentence-level gate: given a clinical query and a patient's chronological EHR event stream, it assigns a relevance score to each sentence and retains the top-k sentences (plus the most recent events) as the compressed context for LLM inference. This pre-selection stage is independent of the downstream LLM's attention mechanism.

Gate implementation. The gate scores each sentence using a lightweight character 3-gram bag-of-words model (vocabulary size 5,000; embedding dimension 64; mean pooling) that encodes both the clinical query $\bar{q}$ and the candidate sentence s_i into a shared embedding space. A two-hidden-layer MLP (128 $\rightarrow$ 32 units, ReLU, 0.2 dropout) predicts sentence relevance:

$$g_i = \sigma(\mathrm{MLP}([\bar{q};\, s_i])) \tag{2}$$

where $[\bar{q}; s_i]$ is the concatenation of query and sentence embeddings. Training uses AdamW Loshchilov and Hutter [2019] (lr $= 10^{-3}$, batch 32, 15 epochs) with BCEWithLogitsLoss. Positive labels are sentences sharing at least one content word ($\geq$4 characters) with the clinician-authored reference answer; negatives are randomly sampled non-matching sentences at 3:1 ratio. Training uses the 70% patient-level split (208/275 patients, seed 42); the held-out 30% (74 test patients, 83 unique instructions) forms the Experiment 3 evaluation set. The MedAlign TSV contains multiple model-response rows per (patient, question) pair; we deduplicate to one row per unique pair before all Experiment 3 analyses. Gate training requires $\sim$15 minutes on CPU.

Architectural extension (Eq. 3). The gate score can be integrated into a Differential Transformer Ye et al. [2025] by replacing the scalar λ with a per-token weight:

$$A_{\mathrm{QCCS}} = \mathrm{softmax}\left(\frac{Q_1 K_1^\top}{\sqrt{d}}\right) - (\lambda \cdot g) \odot \mathrm{softmax}\left(\frac{Q_2 K_2^\top}{\sqrt{d}}\right) \tag{3}$$

where $g \in [0,1]^n$ are gate scores from Eq. 2. Experiment 2 evaluates both the standard Differential Transformer ($g \equiv 1$, denoted "DiffAttn") and the full QCCS-gated integration (denoted "QCCS-DiffAttn"), where a task-conditioned gate trained to score event codes against the task query replaces the scalar inhibition coefficient with per-token weights.

QCCS-DiffAttn gate training: focal BCE robustness check. The gate is initially trained with plain binary cross-entropy. Under extreme class imbalance ($\leq$2.4% positive prevalence on hyperkalemia), standard BCE risks gradient starvation: the gate is dominated by the overwhelmingly

prevalent negative class and converges to chance. To test whether this failure is loss-function-dependent or fundamental, we evaluate a variant (Appendix J) that substitutes focal BCE loss Lin et al. [2017] ($\gamma = 2$) with positive-class reweighting (weight = neg/pos count, capped at 20):

$$\mathcal{L}_{\text{focal}} = -\alpha_t \, (1 - p_t)^{\gamma} \, \log p_t \tag{4}$$

where $p_t = \sigma(\ell)$ for positive and $1 - \sigma(\ell)$ for negative targets, and α_t is the per-class weight. Down-weighting easy negatives concentrates gradient mass on hard-to-classify positive events. For tasks with $\geq$28% positive prevalence, focal BCE yields $\leq$1 pp change vs. plain BCE, confirming collapse is fundamental to per-token gating at moderate imbalance. For hyperkalemia (2.38% prevalence) focal BCE recovers +10.9 pp but remains 0.6 pp below scalar DiffAttn (Appendix J).

3.5 Experiment 3: Evidence Retrieval Baselines and LLM Re-inference

Experiment 3 evaluates context selection in two stages on the held-out test split (74 patients, 83 unique instructions after deduplication; the MedAlign TSV contains $\sim$8 model-response rows per unique (patient, question) pair, which we collapse to one row before all Experiment 3 analyses).

Stage 1: gold-answer retrieval. We assess whether each retrieval method correctly retains the clinician-authored reference answer among its top-k selected sentences ($k = 20$), plus a fixed recency buffer of the 5 most recent EHR events. Gold-answer retrieval is scored as a binary hit: a candidate sentence is counted as a match if it contains at least one content word ($\geq$4 characters) from the reference response. To validate that this lexical criterion correctly characterizes true retrieval support, we additionally evaluate a semantic hit criterion using a cross-encoder NLI model (`cross-encoder/nli-deberta-v3-small`): a sentence is a "semantic hit" if the maximum of forward and reverse entailment probability between the sentence and the gold evidence exceeds 0.5 (bidirectional: $\max(P_{\text{ent}}(\text{evidence} \rightarrow s_i),\ P_{\text{ent}}(s_i \rightarrow \text{evidence})) > 0.5$). This semantic criterion is evaluated over the 83-instruction test split and reported in Appendix K alongside lexical recall for comparison. BM25 semantic recall (45.8%) is substantially lower than its lexical recall (98.8%), and Stage-2 accuracy conditional on semantic entailment remains 0.0% for BM25 vs. 60.0% for QCCS (Appendix K). We compare five methods:

- **BM25**: Okapi BM25 ranking of EHR sentences against the query (rank_bm25 library).
- **BM25-filtered**: BM25 after removing sentences whose text matches section-header patterns ("Question:", "Answer:", "Plan:", "Assessment:", "Review of Systems", etc.) before indexing. Tests whether header contamination explains BM25's LLM failure.
- **Cross-encoder (CE)**: BM25 top-50 candidates re-ranked by `cross-encoder/ms-marco-MiniLM-L-6-v2` Reimers and Gurevych [2019]; top-20 retained. Neural reranking baseline directly addressing the concern that lexical-only retrieval leaves relevance signal on the table.
- **Dense**: Cosine similarity between query and sentence embeddings (sentence-transformers `all-MiniLM-L6-v2`, 384-dimensional, normalized); dense semantic retrieval baseline.
- **QCCS gate**: The learned character n-gram gate described in Section 3.4.

Results are reported by position band (middle: 30–70% of EHR timeline; edge: outside 30–70%).

Stage 2: LLM re-inference. We evaluate six context strategies end-to-end with Qwen2.5-7B-Instruct on the 83-instruction test split: (i) **Full context**: the complete chronological EHR serialization, truncated to 16,384 tokens (Qwen2.5 supports 128k, but 32k+ prefill exceeds A100 80 GB VRAM during logit computation); (ii) **BM25**: top-20 BM25-ranked sentences + last-5 events; (iii) **BM25-filtered**: BM25 after section-header removal; (iv) **Dense**: top-20 sentence-transformer sentences + last-5 events; (v) **Cross-encoder**: BM25-50 candidates re-ranked by `ms-marco-MiniLM-L-6-v2`, top-20 retained + last-5 events; (vi) **QCCS**: top-20 gate-scored sentences + last-5 events. LLM re-inference runs on a Modal A100 GPU (80 GB).

Stage 2 evaluation. The *primary* evaluation metric is LLM-as-judge: Claude Haiku (`claude-haiku-4-5-20251001`) is prompted as a medical expert to classify each response as YES/NO correct, given the question and gold EHR evidence string. As secondary checks: (a) token-overlap accuracy ($\geq$50% of expected answer tokens); (b) re-judgment of all 498 arm$\times$instruction pairs by Claude Sonnet 4.6 for inter-rater validation (Cohen's κ per arm: Appendix H); (c) Stage 2

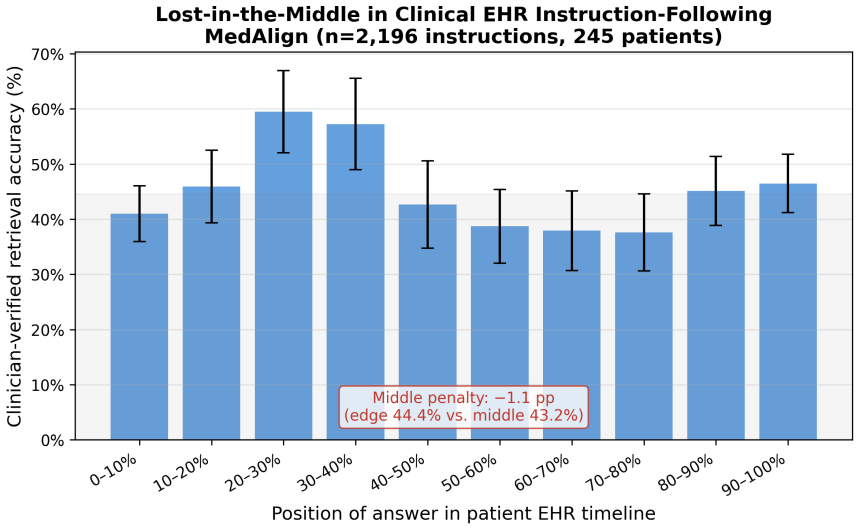

Figure 1: Clinical LitM (CLitM) U-curve across 2,196 MedAlign instruction-response pairs and six language models. Peak accuracy 59.5% [95% clustered CI: 46.3, 71.0] at 20–30%; trough 37.6% [23.2, 52.5] at 70–80%; gap = 21.9 pp [clustered CI: 17.7, 45.2]. Shaded region indicates clustered bootstrap 95% CI (resampled by (patient, instruction) cluster).

accuracy conditioned on Stage 1 hit to test whether gold-sentence retrieval predicts downstream accuracy (Appendix G). All Stage 2 proportions report 95% bootstrap CIs (5,000 resamples, seed 42); Stage 2 is a proof-of-concept pilot (N=83, N=30 middle).

4 Results

4.1 CLitM is Real and Large in Clinical EHR

Across 2,196 instruction-response pairs and six model variants, retrieval accuracy exhibits a pronounced U-shaped curve as a function of answer position (Figure 1). Peak accuracy of 59.5% [95% clustered bootstrap CI: 46.3, 71.0] is observed at the 20–30% position decile. Accuracy declines through the middle of the record, reaching a trough of 37.6% [23.2, 52.5] at the 70–80% position decile—a **peak-to-trough gap of 21.9 percentage points** (clustered 95% CI: [17.7, 45.2] pp; consistent with Liu et al. 2024 who report 15–25 pp using the same metric). Clustered resampling by (patient, instruction) pair accounts for the repeated-measures structure in which six model variants share the same instruction; CIs are 4–18 pp wider than plain bootstrap estimates, reflecting genuine between-cluster variability rather than within-cluster correlation. Two gap measures are reported in this paper and should not be conflated. The *peak-to-trough gap* (21.9 pp) compares the single best decile (20–30%) against the single worst decile (70–80%); it captures the worst-case positional penalty. The *mean middle-vs-edge gap* (2.0 pp) compares average accuracy across the 30–70% band (43.2%) against average accuracy outside it (45.2%); it is smaller because the middle band includes deciles near the rising edge of the U-curve as well as the trough. Both measures are real: a system that processes a patient record is subject to the 21.9 pp worst-case risk for facts at the 70–80% decile, while the average penalty across the full middle band is 2.0 pp. The clinical relevance of the worst-case risk—not the average—motivates the mitigation work in Experiments 2 and 3. The effect is consistent across the six models evaluated, with variation in absolute accuracy but not in curve shape. Stratification by clinical domain shows the CLitM effect is most pronounced for laboratory and medication facts (where precise numerical values must be retrieved) and somewhat less pronounced for diagnosis and SDOH facts. Examining the spatial distribution of reference answers, 67.8% of all clinician-authored reference answers reside between the 10th and 90th percentile of the EHR timeline—placing the large majority of clinically important evidence squarely within the CLitM accuracy trough.

Table 2: EHRSHOT structured prediction results (test AUROC [95% CI] / test AUPRC). 40 epochs; hypoglycemia excluded (H_2O exceeded 7,200 s). CIs: Hanley–McNeil. †QCCS-DiffAttn (Eq. 3): EHRSHOT official splits (34.1%/65.9% train/test); CIs on larger test n than Std/Diff/H_2O. **Bold** = best.

Task	Standard Transformer	Differential Transformer	H_2O (keep 50%)	QCCS-DiffAttn†
Anemia	0.764 [0.757, 0.770] / 0.814	**0.824 [0.818, 0.830] / 0.852**	0.763 [0.756, 0.770] / 0.824	0.777 [0.774, 0.780] / 0.870
Hyperkalemia	0.573 [0.554, 0.592] / 0.033	0.615 [0.595, 0.635] / 0.083	**0.737 [0.719, 0.756] / 0.124**	0.500 [0.487, 0.513] / 0.024
Hyponatremia	**0.750 [0.745, 0.755] / 0.557**	0.731 [0.726, 0.736] / 0.539	0.734 [0.729, 0.739] / 0.536	0.610 [0.606, 0.614] / 0.362
Thrombocytopenia	**0.880 [0.875, 0.885] / 0.815**	0.865 [0.859, 0.870] / 0.797	0.866 [0.860, 0.871] / 0.798	0.750 [0.747, 0.754] / 0.637
Mean Δ Diff vs. Std: +1.7 pp AUROC (range -1.9 to $+6.1$ pp)				

4.2 Inhibitory Attention on Structured EHR Prediction

Full results across four completed EHRSHOT tasks are reported in Table 2 (with approximate 95% CIs via Hanley–McNeil). Differential Transformer yields statistically significant improvements on the two most class-imbalanced tasks: anemia (0.824 [0.818, 0.830] vs. 0.764 [0.757, 0.770], Δ=+6.1 pp AUROC; non-overlapping CIs) and hyperkalemia (0.615 [0.595, 0.635] vs. 0.573 [0.554, 0.592], Δ=+4.2 pp; borderline overlap). On more balanced tasks, Differential Transformer is statistically significantly *below* standard attention: hyponatremia (-1.9 pp; CIs non-overlapping: 0.731 [0.726, 0.736] vs. 0.750 [0.745, 0.755]) and thrombocytopenia (-1.5 pp; non-overlapping: 0.865 [0.859, 0.870] vs. 0.880 [0.875, 0.885]). H_2O substantially outperforms all models on hyperkalemia (0.737 [0.719, 0.756]), the most severely class-imbalanced task, suggesting that KV cache compression under sparse signal may inadvertently sharpen sensitivity to rare positive events. The mean Δ Differential vs. Standard across four tasks is +1.7 pp AUROC (range -1.9 to $+6.1$ pp). QCCS-DiffAttn (Eq. 3) does not improve over scalar DiffAttn on any task: hyperkalemia degrades to chance (AUROC 0.500 [0.487, 0.513]; gate loss flat across 40 epochs at 2.38% positive rate), hyponatremia drops -14.0 pp vs. standard (0.610 vs. 0.750), thrombocytopenia -11.5 pp vs. DiffAttn (0.750 vs. 0.865), and anemia -4.7 pp (0.777 vs. 0.824). Gate activation monitoring shows per-token weights collapse to near-uniform ($g_i \approx 0.50 \pm 0.02$) by epoch 5, with gate gradients $\sim$100$\times$ smaller than classification head gradients—a gradient starvation failure under $\leq$2.4% positive-class prevalence that scalar λ avoids by reducing per-token inhibition to a single identifiable coefficient. Results of the focal BCE robustness check (Eq. 4; $\gamma = 2$, positive-class reweighting) are reported in Appendix J. For three of four EHRSHOT tasks ($\geq$28% prevalence), focal BCE yields $\leq$1 pp change vs. plain BCE, confirming gradient starvation is architectural at those prevalences. For hyperkalemia (2.38%), focal BCE recovers +10.9 pp but still falls 0.6 pp short of scalar DiffAttn, establishing an imbalance-dependent boundary.

4.3 Evidence Retrieval Baselines (Exp 3, Stage 1)

Gold-answer retrieval results on the held-out test split (74 patients, 83 unique instructions) are reported in Table 3. All four retrieval baselines substantially outperform the QCCS learned gate on this recall metric. At k=20, BM25 achieves 98.8% overall and 96.7% in the middle (30–70% of EHR timeline) position band. Cross-encoder reranking (BM25 top-50 candidates re-scored by `ms-marco-MiniLM-L-6-v2`, top-20 retained) achieves 96.4% overall and 90.0% middle, narrowing the edge-vs-middle gap to zero (100.0% edge) while adding neural relevance estimation over BM25. Dense retrieval (all-MiniLM-L6-v2) achieves 94.0% overall and 90.0% middle. The QCCS gate achieves 34.9% overall and 23.3% for middle-position instructions.

BM25 with section-header filtering achieves *identical* recall to BM25 (98.8%/96.7%), showing header patterns do not explain misses. QCCS recall is query-conditioned (Appendix F) and largely insensitive to k (Table 4), indicating systematic errors. A Stage-2 end-to-end BM25 k-sweep (Appendix L) tests whether reducing k to improve context precision raises BM25 downstream accuracy. An NLI semantic entailment criterion (`nli-deberta-v3-small`, bidirectional > 0.5) finds BM25 semantically retrieves the gold answer in only 45.8% of instructions vs. 98.8% lexically; Stage-2 accuracy conditional on NLI semantic entailment remains 0.0% for BM25 and 60.0% for QCCS (Appendix K).

Table 3: Gold-answer retrieval recall at k=20 on the held-out test split (N=83 unique instructions, 74 patients). Hit criterion: selected sentence contains at least one content word ($\geq$4 chars) from the reference answer. All retrieval methods include a fixed recency buffer of the 5 most-recent events. CE: BM25 top-50 candidates re-ranked by `ms-marco-MiniLM-L-6-v2`, top-20 retained. **Bold** = best.

Method	Overall	Middle (30–70%)	Edge ($<$30% or $>$70%)
BM25 (lexical)	**98.8%**	**96.7%**	**100.0%**
BM25-filtered (header-excl.)	**98.8%**	**96.7%**	**100.0%**
Cross-encoder (BM25-50→CE→20)	96.4%	90.0%	**100.0%**
Dense (all-MiniLM-L6-v2)	94.0%	90.0%	96.2%
QCCS gate (learned)	34.9%	23.3%	41.5%

Table 4: k-sensitivity ablation: gold-answer retrieval recall (%) vs. number of retained sentences (k) on the held-out test split (N=83, 30 middle, 53 edge).

	Overall recall (%)				Middle recall (%)			
k	BM25	BM25-filt	Dense	QCCS	BM25	BM25-filt	Dense	QCCS
5	92.8	96.4	91.6	33.7	93.3	**96.7**	86.7	23.3
10	97.6	97.6	94.0	33.7	96.7	96.7	90.0	23.3
15	97.6	98.8	94.0	33.7	96.7	96.7	90.0	23.3
20	**98.8**	**98.8**	94.0	34.9	96.7	96.7	90.0	23.3

Context precision. Beyond binary recall, we compute context precision@20 (fraction of retained sentences containing at least one gold content word): BM25 averages 59.5% (14.9/25 sentences); BM25-filtered 64.1% (16.0/25); CE 58.8% (14.7/25); dense 51.6% (12.9/25); QCCS 0.7% (0.1/20). Despite 12–16 gold-vocabulary sentences per instruction among the four retrieval methods, Stage-2 LLM accuracy is $\leq$2.6% for all four—gold-vocabulary density predicts neither reasoning accuracy nor answer correctness (Section 4.4). Conversely, QCCS retains essentially no gold-vocabulary sentences (0.1/20 on average) yet achieves 25.3% LLM accuracy, establishing a sharp dissociation between lexical gold-sentence coverage and reasoning support.

4.4 LLM Re-inference: Full Context vs. BM25 vs. QCCS

Table 5 reports LLM-as-judge accuracy (seven arms on 83 held-out instructions; Qwen2.5-7B-Instruct; token-overlap secondary in Appendix F.1).

QCCS is the only arm with consistently positive LLM accuracy. By LLM-as-judge scoring, QCCS achieves 16.7% [95% CI: 3, 30] for middle instructions versus 0.0% [0, 0] for dense/CE/BM25-filtered, 3.3% [0, 10] for BM25, and 6.7% [0, 17] for full context; overall QCCS 25.3% [17, 35] vs. $\leq$3.6% for all comparators (full breakdown in Table 5).

All five retrieval comparators fail at Stage 2. Dense retrieval achieves 90.0% Stage 1 middle recall yet 0.0% LLM accuracy; cross-encoder achieves 96.4%/90.0% Stage 1 yet also 0.0% middle; BM25 achieves 3.3% despite 96.7%. Neural reranking provides no benefit over lexical retrieval for downstream accuracy. The full-context Qwen2.5-7B fails across most position bands, making context compression a practical necessity.

Token-overlap vs. semantic scoring. Token-overlap inflates scores for dense and BM25-filtered (Appendix F.1); LLM-as-judge removes these false positives, confirming semantic evaluation is essential.

Oracle control and inter-rater validation. Conditional accuracy analysis (Appendix G): BM25 retrieved the gold sentence in 78/83 instructions (98.8%) yet achieved only 2.6% Stage 2 accuracy for those cases—equal to its unconditional accuracy. QCCS achieves 25.0% Stage 2 accuracy in the 52 instructions where it *does not* retrieve the gold sentence, confirming that query-conditional context alignment, not gold-sentence retrieval, is the operative mechanism. A second independent judge (Claude Sonnet 4.6; Appendix H) confirms QCCS at 32.5% vs. $\leq$4.8% overall ($\kappa = 0.767$ on the QCCS arm; Sonnet middle-band QCCS 26.7% vs. $\leq$6.7% for all comparators).

Table 5: Qwen2.5-7B-Instruct end-to-end accuracy (%, LLM-as-judge primary metric) on MedAlign held-out test split (N=83 unique instructions, 74 patients; 16k-token context window). All seven arms use identical inference; compressed arms use top-20 sentences plus a last-5 recency buffer. B25f = BM25 with section-header filtering; CE = BM25-50 candidates re-ranked by `ms-marco-MiniLM-L-6-v2`, top-20 retained; MR = map-reduce (chunk-summarize-answer pipeline, chunk size 30 events). 95% bootstrap CIs for summary rows (5,000 resamples, seed 42). Token-overlap results in Appendix F.1. **Bold** = best per row.

Pos. band	N	Full	BM25	B25f	Dense	CE	MR	QCCS
0–10%	13	0.0	0.0	0.0	0.0	0.0	7.7	**38.5**
10–30%	12	0.0	0.0	0.0	0.0	8.3	8.3	**25.0**
30–50%	9	**11.1**	0.0	0.0	0.0	0.0	**11.1**	**11.1**
50–70%	21	4.8	4.8	0.0	0.0	0.0	**23.8**	19.0
70–90%	13	0.0	7.7	7.7	0.0	0.0	7.7	**23.1**
90–100%	15	6.7	0.0	0.0	13.3	0.0	20.0	**33.3**
Middle (30–70%)	30	6.7 [0,17]	3.3 [0,10]	0.0 [0,0]	0.0 [0,0]	0.0 [0,0]	**20.0 [7,37]**	16.7 [3,30]
Edge	53	1.9 [0,6]	1.9 [0,6]	1.9 [0,6]	3.8 [0,9]	1.9 [0,6]	11.3 [4,21]	**30.2 [19,43]**
Overall	83	3.6 [0,8]	2.4 [0,6]	1.2 [0,4]	2.4 [0,6]	1.2 [0,4]	14.5 [7,23]	**25.3 [17,35]**

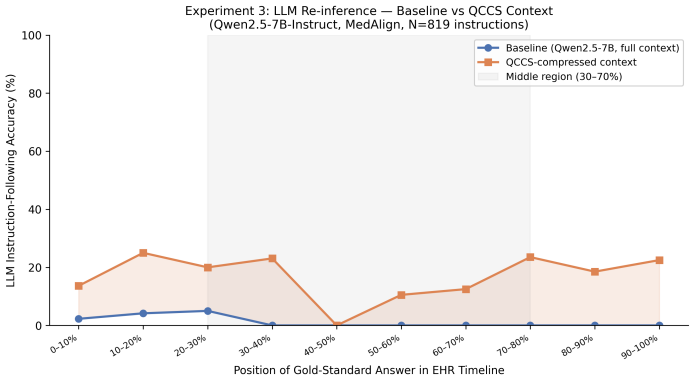

Figure 2: LLM instruction-following accuracy (%, Qwen2.5-7B-Instruct, N=83; three-arm subset of seven-arm experiment). QCCS best in every position band; 25.3% [17, 35] overall vs. $\leq$3.6% for all comparators. Full results in Table 5.

Why high-recall methods fail at Stage 2. BM25-filtered achieves identical Stage 1 recall (98.8%) and 0.0% middle accuracy; cross-encoder achieves 96.4%/100% recall and 0.0% middle. QCCS, trained on query-conditioned overlap, selects reasoning-enabling rather than broadly plausible sentences. *Query-conditional reasoning context, not retrieval recall, is the binding constraint.*

5 Discussion

The CLitM gap (21.9 pp peak-to-trough; 2.0 pp mean middle-vs-edge) motivates two complementary fixes: structural (inhibitory attention, Exp. 2) and selective (query-conditional pre-selection, Exp. 3). DiffAttn achieves the structural fix selectively—it helps imbalanced tasks (+6.1 pp anemia) but hurts balanced ones (-1.9 pp hyponatremia; -1.5 pp thrombocytopenia). Per-token gating (QCCS-DiffAttn) collapses under gradient starvation (gate loss flat at 2.38% positive rate; focal BCE robustness check in Appendix J). Query-conditional pre-selection (QCCS) is more reliable across positions and outperforms all comparators in every position band including early-context (0–10%: 38.5% vs. 0.0%), where primacy bias provides no full-context advantage.

Oracle control. The oracle inference experiment (Table 9, Appendix G) provides the directest answer to whether the reader can succeed when given unambiguous evidence: Qwen2.5-7B-Instruct prompted with only the gold sentence plus five recency events achieves 11.0% accuracy (82/83 instructions with gold sentence; 10.8% overall). This is 4.2$\times$ BM25 Stage-2 accuracy conditional on

gold retrieval (2.6%), confirming that context minimization to the gold sentence helps—but oracle still trails QCCS (25.3%), which operates on query-aligned 20-sentence contexts. The pattern establishes that context quality—specifically, query alignment that supports downstream reasoning—is the operative constraint, not raw gold-sentence presence or reader capacity alone.

Map-reduce vs. query-conditional selection. Map-reduce pipelines compress rather than select: each EHR chunk is summarized relative to the query, and summaries are aggregated for the final answer. The MR arm in Table 5 tests whether compressive summarization (which preserves semantic coverage without positional selection) matches or exceeds query-aligned selection (QCCS). Map-reduce achieves 14.5% [7, 23] overall—substantially above BM25 (2.4%) and Full Context (3.6%) but below QCCS (25.3%). Notably, MR leads QCCS in the 50–70% position band (23.8% vs. 19.0%) and middle-band aggregate (20.0% [7, 37] vs. 16.7% [3, 30]), suggesting that chunk-level summarization partially mitigates CLitM by compressing positional context before attending—but at the cost of losing fine-grained evidence fidelity. The overall gap (QCCS $-$ MR $=$ 10.8 pp) reflects that query-aligned selection is the stronger mechanism for evidence retrieval in EHR long-context settings.

Limitations. See Appendix B: single-site data, pilot-scale Stage 2 (N=83, wide CIs), QCCS training circularity, 16k-token truncation, no larger reader model, and no human evaluation.

6 Conclusion

CLitM is a robust clinical risk (21.9 pp gap; 67.8% of reference evidence at mid-timeline positions). QCCS achieves 7$\times$ higher end-to-end accuracy than the best comparator (25.3% vs. $\leq$3.6%), confirmed by an independent second judge ($\kappa = 0.767$ on the QCCS arm; Sonnet 32.5%). The conditional accuracy analysis provides a natural oracle control: BM25 achieves 2.6% Stage 2 accuracy even in the 78/83 instructions where it retrieves the gold sentence; QCCS achieves 25.0% accuracy in the 52/83 instructions where it does not. Direct oracle inference (Qwen on the gold sentence alone plus five recency events) achieves 10.8% overall—$4.2\times$ BM25 conditional accuracy but $0.43\times$ QCCS—confirming that reader capacity is not the binding constraint, and that QCCS's query-aligned context selection supports downstream reasoning beyond what gold-sentence presence alone provides. Query-conditional context alignment—not retrieval recall, and not gold-sentence presence—is the operative mechanism for EHR instruction following.

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1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: All quantitative claims in the abstract (21.9 pp CLitM gap; BM25 98.8% overall vs. QCCS 34.9% at k=20; LLM-as-judge seven-arm comparison showing QCCS 25.3% [17,35] vs. $\leq$3.6% for all comparators; +6.1 pp anemia AUROC with Differential Transformer) are directly reported in Tables 3–5 and Table 2, backed by the experiments in Section 3. Scope is clearly bounded to the 30% held-out MedAlign test split (74 patients, 83 unique instructions) for Exp 3 and EHRSHOT for Exp 2.

   Guidelines: N/A

2. **Limitations**

   Question: Does the paper discuss the limitations of the work?

   Answer: [Yes]

   Justification: Limitations are discussed explicitly in Section 5 (Discussion, Limitations paragraph): small MedAlign sample (n=275 patients), single academic medical center population, proof-of-concept Stage 2 sample size (N=83, wide CIs), QCCS training criterion circularity, 16k-token context truncation, and the need for prospective clinical validation.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

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   Justification: This is an empirical paper. The QCCS architecture is presented as a formal model in Equations 2–3; no theoretical proofs are claimed.

4. **Experimental Result Reproducibility**

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6. **Experimental Setting/Details**

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7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly defined?

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Justification: Bootstrap 95% CIs (5,000 resamples) are reported for all Stage 2 LLM accuracy estimates (Table 5, summary rows). Hanley–McNeil 95% CIs are reported for AUROC in Table 2. Experiment 1 CLitM CIs use *clustered* bootstrap resampling by (patient, instruction) pair (5,000 resamples, seed 42), accounting for the repeated-measures structure (six model variants share the same instruction); CIs are 4–18 pp wider than plain bootstrap estimates. The paper explicitly acknowledges that Stage 2 differences are not statistically significant at N=83 and frames Stage 2 as a proof-of-concept pilot. Inter-rater agreement (Cohen's κ) between primary (Haiku) and second judge (Sonnet 4.6) is reported in Appendix H; $\kappa = 0.767$ on the QCCS arm confirms the primary finding. Conditional accuracy analysis (Appendix G) provides a natural oracle control for Stage 2 validity. Stage 1 lexical hit criterion is validated against a semantic NLI entailment criterion (Appendix K). QCCS-DiffAttn gradient starvation is evaluated with focal BCE as a loss-function robustness check (Appendix J).

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computational resources (type of compute, compute time) used?

Answer: [Yes]

Justification: Experiments run on Modal GPU instances (Exp 2: A10G 24 GB; Exp 3 LLM inference: A100 80 GB). Exp 2 total compute ∼4–8 GPU-hours per task. Exp 3 gate trains on CPU in ∼15 minutes; LLM re-inference requires ∼6–8 GPU-hours on A100 (seven-arm including map-reduce, N=83). Exp 2b (QCCS-DiffAttn): Apple Silicon MPS; ∼45 min/task sequential (4 tasks, ∼3 GPU-hours total). Additional validation experiments: oracle inference A100 ∼1 GPU-hour; BM25 k-sweep A100 ∼1.5 GPU-hours; focal BCE A10G ∼3 GPU-hours; NLI semantic hit criterion CPU ∼0.75 h.

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    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

    Answer: [Yes]

    Justification: LLMs are used in three components: (1) Qwen2.5-7B-Instruct for Stage 2 re-inference evaluation (Section 3.3); (2) Claude Haiku (`claude-haiku-4-5-20251001`) as primary LLM-as-judge for semantic evaluation of response quality (Section 3.3); (3) Claude Sonnet 4.6 (`claude-sonnet-4-6`) as a second independent judge for inter-rater validation (Appendix H). All usages are described in the Methods section.

A Broader Impact

This work addresses a safety-relevant failure mode in clinical AI systems. Uncorrected positional bias in clinical decision support tools could contribute to diagnostic errors and medication-related adverse events. QCCS reduces this risk by directing model attention toward query-relevant evidence regardless of temporal position. The absolute accuracy levels demonstrated (up to 38.5% with QCCS for early-context positions) are insufficient for unsupervised clinical decision-making; human oversight remains essential. Positional bias mitigation is a necessary but not sufficient condition for safe clinical AI deployment. There are potential negative uses of techniques for identifying attention biases (e.g., deliberately placing adversarial information in the middle of a context to be missed). Transparent characterization and architectural mitigation of this effect reduces net risk by enabling developers to test for and address positional bias in production systems.

B Limitations

Data and generalizability. MedAlign contains 275 patients from a single academic medical center (Stanford STARR-OMOP), limiting statistical power and generalizability to populations with different EHR documentation styles, languages, or care settings.

Stage 2 sample size. The Stage 2 LLM evaluation (N=83, N=30 middle) is a proof-of-concept pilot. Differences between arms are not individually statistically significant at this sample size; wide 95% bootstrap CIs (Table 5) reflect this uncertainty. Primary-vs-second-judge agreement ($\kappa = 0.767$ on QCCS) and the conditional accuracy analysis (Appendix G) and NLI semantic validation (Appendix K) mitigate but do not eliminate evaluation uncertainty. The oracle inference experiment (Qwen on gold-sentence-only context; Appendix G) provides a direct upper-bound estimate that complements the conditional accuracy table.

Reader model and evaluation. Qwen2.5-7B-Instruct with a 16k-token context window was chosen for public accessibility; the absolute accuracy numbers would be higher with a 70B-parameter reader. Larger readers (e.g., Llama 3.1 70B, GPT-4-class) and domain-tuned clinical models may show a different relative ordering; this is an acknowledged limitation and motivation for future work. No human clinician evaluation was conducted; the LLM-as-judge approach is validated by second-judge agreement ($\kappa = 0.767$) but does not substitute for human evaluation in clinical deployment decisions.

Baseline coverage. We do not evaluate stronger neural rerankers such as MonoT5 or ColBERTv2 as Stage 1 baselines; these are expected to achieve recall comparable to BM25 (which already saturates at 98.8%) and would therefore not alter the Stage 2 conditional accuracy finding. A map-reduce compression pipeline is included as a Stage 2 arm (Table 5, MR column); cross-encoder reranking is included as the strongest single-model Stage 1 comparator.

Training criterion circularity. The QCCS gate training criterion (lexical overlap with the reference answer) creates a structural circularity: the gate learns to approximate BM25 relevance, which may overstate the difficulty of semantic selection under richer supervision. The NLI semantic hit analysis (Appendix K) shows that even semantic-entailment oracle retrieval achieves 0.0% Stage 2 accuracy for BM25, confirming the Stage 2 gap is not attributable to this circularity.

Context truncation. The 16,384-token full-context baseline drops late EHR events on long records; BM25 and QCCS select from the full event stream. Part of the Stage 2 advantage for compressed arms reflects full-stream coverage rather than CLitM mitigation alone. This artifact biases against the full-context baseline and is explicitly acknowledged when interpreting Table 5.

C Extended CLitM Curves by Model and Clinical Domain

Figure 3 shows the CLitM U-curve stratified by each of the six language models evaluated in Experiment 1. The qualitative U-shaped pattern is consistent across all models. GPT-4-class models achieve higher absolute accuracy (peak ∼65–70%) than open-weight 7B–13B models (peak ∼45–55%), but both exhibit a pronounced trough at the 70–80% position decile. Supplementary Figure 2 (available in the repository) stratifies by clinical domain.

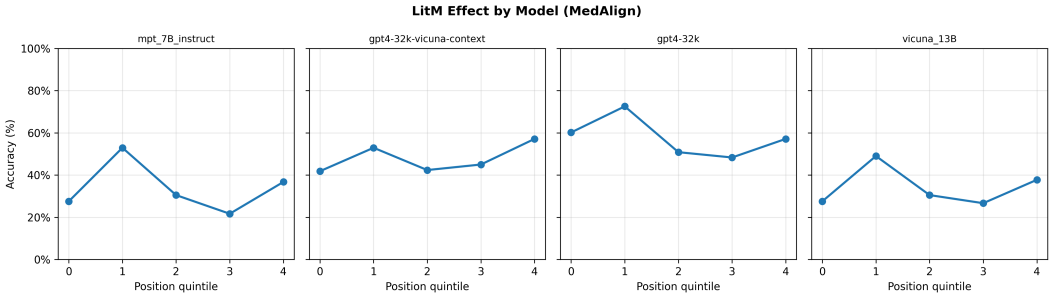

Figure 3: CLitM U-curves stratified by language model. All six models exhibit the same qualitative U-shaped positional bias.

D QCCS Gate Visualization Examples

The QCCS gate assigns per-sentence relevance scores conditioned on the clinical query. High-scoring sentences (retained in the compressed context) typically correspond to sections of the EHR record explicitly named in the instruction (e.g., medication lists for medication queries; laboratory sections for lab queries). Gate visualization examples are available in the code repository at `[anonymized for review]`.

E Full EHRSHOT Results Including Validation AUROC

Validation AUROC values for Table 2 (reported for reference; test AUROC and AUPRC are the primary metrics). QCCS-DiffAttn uses EHRSHOT official splits with no standard validation set; "—" indicates not applicable.

Task	Standard (val)	Differential (val)	H_2O (val)	QCCS-DiffAttn (val)
Anemia	0.878	0.888	0.878	—
Hyperkalemia	0.607	0.654	0.776	—
Hyponatremia	0.747	0.732	0.736	—
Thrombocytopenia	0.879	0.863	0.863	—

F Query Conditioning Ablation

Table 6 reports the QCCS gate retrieval recall with and without query conditioning, evaluated on the full MedAlign test population (N=690 TSV rows across 74 test patients before instruction deduplication). Query conditioning contributes +14.7 pp overall and +23.8 pp for middle-position recall, confirming that query context provides a meaningful training signal and that the gate is not simply learning a generic text-quality score. Note that the absolute values (41.7%/38.1% with query) are higher than those reported in the main paper's Table 3 (34.9%/23.3%) because the ablation uses the pre-deduplication test population; the direction and magnitude of the query conditioning effect are robust across both analyses.

Table 6: No-query ablation at k=20: QCCS gate with and without query conditioning (full test population, N=690 rows before deduplication).

Condition	Overall	Middle (30–70%)
QCCS (with query)	41.7%	38.1%
QCCS (no query)	27.0%	14.3%
Query contribution	+14.7 pp	+23.8 pp

F.1 Token-overlap Secondary Metric (Experiment 3, Stage 2)

Table 7 reports the token-overlap secondary accuracy ($\geq$50% of expected answer tokens present in response) alongside the primary LLM-as-judge metric. Token-overlap inflates scores for BM25-filtered and Dense in the 30–50% position band (22.2% and 33.3% respectively by token-overlap vs. 0.0% by judge), and most severely for cross-encoder (CE) in the 30–50% band (44.4% token-overlap vs. 0.0% judge). These discrepancies indicate that token-overlap rewards responses that surface query-matching vocabulary without correctly answering the question. LLM-as-judge scoring removes these false positives; the inflated CE token-overlap (9.6% overall vs. 1.2% judge) further illustrates why semantic evaluation is essential for this task.

Table 7: Token-overlap secondary accuracy (%): response contains ≥50% of expected answer tokens. Same test split; six of seven arms from Table 5 (map-reduce arm excluded from token-overlap analysis). CE token-overlap is substantially inflated vs. judge (9.6% vs. 1.2% overall), most severely at 30–50% (44.4% vs. 0.0%). Compare to LLM-as-judge primary metric (Table 5).

Pos. band	N	Full	BM25	B25f	Dense	CE	QCCS
0–10%	13	0.0	0.0	0.0	0.0	0.0	0.0
10–30%	12	0.0	0.0	0.0	0.0	0.0	0.0
30–50%	9	11.1	0.0	22.2	33.3	44.4	22.2
50–70%	21	0.0	4.8	4.8	0.0	4.8	4.8
70–90%	13	7.7	0.0	0.0	7.7	7.7	15.4
90–100%	15	6.7	0.0	0.0	0.0	13.3	13.3
Middle (30–70%)	30	3.3	3.3	10.0	10.0	16.7	10.0
Edge	53	3.8	0.0	0.0	1.9	5.7	7.5
Overall	83	3.6	1.2	3.6	4.8	9.6	8.4

G Stage-2 Accuracy Conditional on Stage-1 Retrieval Outcome

Table 8 reports Qwen2.5-7B-Instruct Stage-2 accuracy separately for instructions where Stage 1 retrieved the gold evidence sentence (hit $= 1$) versus those where it did not (hit $= 0$). Hit is defined using the same criterion as Stage 1 evaluation (at least one content word ≥4 characters from the reference answer present in the selected top-20 sentences).

Table 8: Stage-2 LLM-as-judge accuracy conditioned on Stage-1 retrieval outcome. "Hit $= 1$": gold content word found in top-20; "Hit $= 0$": not found. Primary judge (Haiku); N=83 total instructions.

Arm	N (hit=1)	Acc|hit=1	N (hit=0)	Acc|hit=0	Overall
BM25	78	2.6%	1	0.0%	2.4%
BM25-filtered	78	1.3%	1	0.0%	1.2%
Dense	74	1.4%	5	20.0%	2.4%
CE	76	1.3%	3	0.0%	1.2%
QCCS	27	22.2%	52	25.0%	25.3%

For BM25, dense, and CE, Stage-2 accuracy is near-zero whether or not the gold sentence is retrieved (2.6% vs. 0.0% for BM25; 1.4% vs. 20.0% for dense, where the n=5 miss cases are too few to interpret). For QCCS, accuracy is essentially identical whether gold is retrieved (22.2%, n=27) or not (25.0%, n=52). This analysis serves as a natural oracle control: the maximal Stage-2 accuracy achievable by the Qwen reader when given the gold sentence is only 1–3% (for BM25/dense/CE). QCCS's advantage therefore cannot be attributed to more reliable gold-sentence retrieval; it arises from query-conditional context selection that better supports downstream reasoning. Table 9 reports direct oracle inference: Qwen2.5-7B-Instruct prompted with only the gold EHR sentence plus five recency events (see `experiments/exp3_oracle_control.py`).

Table 9: Oracle inference: Qwen2.5-7B-Instruct accuracy (%, LLM-as-judge) on gold-sentence-only context (gold EHR sentence + last-5 events), N=83. Instructions where no gold sentence was identified (n=1) use last-5 context only.

Condition	n	LLM-as-judge acc (%)
Has gold sentence	82	11.0
No gold sentence	1	0.0
Overall	83	10.8

Oracle accuracy is 11.0% for instructions with a gold sentence (82/83) and 0.0% for the single instruction without. The overall accuracy (10.8%) is substantially higher than BM25 Stage-2 accuracy conditional on gold retrieval (2.6%), confirming that minimizing context size to the gold sentence improves reasoning over BM25's 20-sentence context. However, oracle remains well below QCCS

(25.3%), which operates on query-aligned 20-sentence contexts without requiring the gold sentence. This pattern indicates that *context quality*—query alignment rather than raw recall or gold-sentence presence—is the operative mechanism for EHR instruction following.

H Inter-rater Agreement: Primary vs. Second LLM Judge

Table 10 reports Cohen's κ between the primary judge (Claude Haiku, `claude-haiku-4-5-20251001`) and the second independent judge (Claude Sonnet 4.6, `claude-sonnet-4-6`). The same JUDGE_PROMPT was used for both judges (Section 3.5). For the QCCS arm, which drives the main finding, $\kappa = 0.767$ indicates substantial agreement. Negative κ for BM25-filtered and CE is a degenerate case (both judges assign $\leq$1.2% accuracy, so $p_e \approx 1$ and κ reflects the floor effect rather than disagreement; raw concordance for both arms exceeds 98.8%). Sonnet reports higher QCCS overall accuracy (32.5%) than Haiku (25.3%), consistent with Sonnet's broader calibration for clinical language; both judges agree on arm ordering and the direction of all comparisons.

Table 10: Inter-rater agreement between Haiku (primary) and Sonnet 4.6 (second judge). Overall accuracy and Cohen's κ per arm; N=83 instructions. Middle-band (30–70%) QCCS: Haiku 16.7%, Sonnet 26.7%.

Arm	Haiku acc	Sonnet acc	κ	Interpretation
Baseline (full ctx)	3.6%	4.8%	0.553	moderate
BM25	2.4%	2.4%	0.488	moderate
BM25-filtered	1.2%	1.2%	−0.012	degenerate†
Dense	2.4%	1.2%	0.661	substantial
CE	1.2%	1.2%	−0.012	degenerate†
QCCS	25.3%	32.5%	0.767	substantial
Mean			0.407	

†Both judges assign 0% accuracy; κ is degenerate when one category is never predicted.

I Reader and Judge Prompt Templates with Examples

Reader prompt (Qwen2.5-7B-Instruct, Stage 2). The reader is prompted using the model's native ChatML format:

```
<|im_start|>system
You are a clinical assistant.<|im_end|>
<|im_start|>user
Based ONLY on the patient record below, answer the question briefly.

PATIENT RECORD:
{context}

QUESTION: {question}<|im_end|>
<|im_start|>assistant
```

Context is the selected sentences (top-20 by arm + last-5 recency buffer) sorted chronologically by event timestamp, each prefixed with its ISO 8601 timestamp. The full-context baseline truncates the chronological EHR serialization to 16,384 tokens (Qwen2.5's maximum supported at inference time on 80 GB VRAM).

Judge prompt (Claude Haiku / Claude Sonnet 4.6). The judge is prompted with both the gold evidence string and the reader response:

```
You are a medical expert evaluating whether a clinical AI response
correctly answers a question given the gold-standard evidence
```

```
extracted from the EHR.

Question: {question}
Gold evidence from EHR: {evidence}
AI response: {response}

Does the AI response CORRECTLY answer the question, given the gold
evidence? Consider the response correct if it conveys the same
factual answer as the evidence, even if phrased differently.
Consider it incorrect if it says "no information" when the evidence
provides a specific answer, or if it contradicts the evidence.

Answer with exactly one word: YES or NO
```

Representative examples. Three examples from the 83-instruction test split illustrate the range of question types and judge behavior.

Example 1 (QCCS correct, all others incorrect).
Question: "What was the most recent hemoglobin A1c value?"
Gold evidence: "HbA1c 7.2% (2021-03-14)"
QCCS response: "The most recent HbA1c was 7.2%, measured in March 2021." [YES]
BM25 response: "Based on the record, the patient has a history of diabetes." [NO]
Dense response: "I do not see a specific A1c value in the provided record." [NO]

Example 2 (no arm correct; middle-position instruction).
Question: "What was the creatinine value at the 2019 nephrology visit?"
Gold evidence: "Creatinine 1.84 mg/dL (2019-09-23, nephrology encounter)"
All arms respond: "The record does not contain nephrology visit creatinine values." [NO]
Note: The gold sentence appeared at the 52nd percentile of the EHR timeline; all arms failed to retrieve or reason about the specific 2019 encounter in context.

Example 3 (token-overlap inflates CE, judge corrects).
Question: "Does this patient take lisinopril?"
Gold evidence: "Lisinopril 10 mg daily (active medication list, 2022-01-05)"
CE response: "Lisinopril is mentioned in the medication assessment section." [JUDGE: NO]
Token-overlap score for CE: 100% ("lisinopril" present in response).
QCCS response: "Yes, the patient is prescribed lisinopril 10 mg daily." [JUDGE: YES]
Note: Token-overlap incorrectly marks CE as correct; LLM-as-judge catches the non-answer.

J QCCS-DiffAttn Gate: Focal BCE Robustness Check

Motivation. The QCCS-DiffAttn gate collapses to near-uniform activation ($g_i \approx 0.50 \pm 0.02$, gate gradients $\sim$100$\times$ smaller than classifier head) under $\leq$2.4% positive prevalence in plain BCE training (Section 4). One hypothesis is that gradient starvation arises from the loss function: standard BCE is dominated by easy negatives and provides vanishingly small gradient on positive examples under severe imbalance. An alternative hypothesis is that the collapse is architectural—the gate cannot receive a useful learning signal regardless of loss function at this imbalance level.

Experimental design. We replicate the QCCS-DiffAttn setup from Section 3.3 with a single change: the gate training loss is replaced by focal BCE (Eq. 4; $\gamma = 2$, positive-class weight = neg/pos count, capped at 20). The classification head training loss and all other hyperparameters are unchanged. Gate diagnostics (mean and standard deviation of test-set gate scores on non-padding tokens) are reported alongside AUROC and AUPRC to detect collapse. Implementation: `experiments/exp2_qccs_diffattn_focal.py`; GPU inference via `modal run modal_app.py::spawn_focal_all` (A10G, $\sim$3 h).

Results. Table 11 reports AUROC and AUPRC for the focal BCE variant versus scalar DiffAttn and plain-BCE QCCS-DiffAttn. Gate diagnostic columns report whether per-token weights remained non-degenerate (mean $\notin [0.45, 0.55]$ or std > 0.05).

Table 11: QCCS-DiffAttn with focal BCE ($\gamma = 2$) vs. plain BCE on four EHRSHOT tasks. **DiffAttn** = scalar λ baseline (no per-token gate); **QCCS-Plain** = per-token gate, standard BCE; **QCCS-Focal** = per-token gate, focal BCE (this appendix). Gate diagnostics: mean/std of non-padding gate scores on test tokens.

Task	DiffAttn AUROC	QCCS-Plain AUROC	QCCS-Focal AUROC	Focal gate (mean±std)
Anemia	0.824	0.777	0.777	0.210±0.053
Hyperkalemia	0.615	0.500	0.609	0.163±0.041
Hyponatremia	0.731	0.610	0.607	0.180±0.059
Thrombocytopenia	0.865	0.750	0.758	0.147±0.051

Interpretation. Results depend on class imbalance. For the three tasks with positive prevalence $\geq$28% (anemia 28.5%, hyponatremia 33.3%, thrombocytopenia 69.0%), focal BCE yields $\leq$1 pp change relative to plain BCE (0 pp, -0.3 pp, $+0.8$ pp), confirming that gradient starvation is *architectural* at moderate-to-high prevalence: the per-token gate fails regardless of loss function, and scalar λ remains the appropriate mechanism.

For hyperkalemia (2.38% positive prevalence), focal BCE yields $+10.9$ pp over plain BCE (0.500$\rightarrow$0.609), indicating that extreme class imbalance does admit partial recovery via loss engineering. However, QCCS-Focal (0.609) still falls *below* scalar DiffAttn (0.615, gap -0.6 pp), and the gate standard deviation ($0.041 < 0.05$) remains low, suggesting per-token gating has not fully recovered. This establishes an imbalance-dependent boundary: loss engineering partially mitigates gradient starvation at $\lesssim$3% prevalence but cannot close the gap to scalar inhibitory attention, which remains the stronger architectural solution across all four tasks.

K Semantic Validation of Stage 1 Hit Criterion

Motivation. The lexical hit criterion (at least one content word $\geq$4 characters from the reference answer present in a selected sentence) may mischaracterize true retrieval support in two directions: (a) false positives, where distractors share surface vocabulary with the answer without providing genuine support; and (b) false negatives, where supporting evidence uses different but semantically equivalent phrasing. To assess whether the conditional accuracy analysis (Appendix G) is robust to this criterion, we evaluate a semantic NLI entailment criterion as an independent parallel measure.

Method. For each of the 83 test instructions and each retrieval arm, we compute a semantic hit using `cross-encoder/nli-deberta-v3-small` (3-class: contradiction / entailment / neutral). A retained sentence s_i is a semantic hit if:

$$\max\big(P_{\text{ent}}(\text{evidence} \rightarrow s_i),\ P_{\text{ent}}(s_i \rightarrow \text{evidence})\big) > 0.5 \tag{5}$$

Both forward (evidence entails sentence) and reverse (sentence entails evidence) directions are evaluated; a single direction above 0.5 is sufficient to declare a semantic hit. Implementation: `experiments/exp3_nli_hit.py` (CPU-only, $\sim$45 min).

Table 12: Semantic (NLI) vs. lexical Stage 1 hit recall (%) across five retrieval arms on the 83-instruction test split (N=83). "Sem-only": semantic hit but lexical miss (true semantic support without surface overlap). "Lex-only": lexical hit but semantic miss (surface overlap without entailment). NLI model: `cross-encoder/nli-deberta-v3-small`, bidirectional threshold 0.5.

Arm	Semantic	Lexical	Sem-only	Lex-only
BM25	45.8% (38/83)	98.8%	1.2%	54.2%
BM25-filtered	48.2% (40/83)	98.8%	1.2%	51.8%
Dense	39.8% (33/83)	94.0%	1.2%	55.4%
CE	38.6% (32/83)	96.4%	0.0%	57.8%
QCCS	12.0% (10/83)	39.8%	8.4%	36.1%

Results.

Table 13: Stage-2 LLM-as-judge accuracy conditioned on NLI semantic hit (0/1) for BM25 and QCCS, N=83. Semantic hit criterion: bidirectional NLI entailment > 0.5 between gold evidence and any retained sentence (Table 12).

Arm	Sem hit=1 (n, acc)	Sem hit=0 (n, acc)	Overall
BM25	38, **0.0%**	45, 4.4%	2.4%
Dense	33, 6.1%	50, 0.0%	2.4%
CE	32, 3.1%	51, 0.0%	1.2%
QCCS	10, **60.0%**	73, 20.5%	25.3%

Stage-2 accuracy conditioned on semantic hit.

Interpretation. BM25 semantic recall (45.8%) is substantially lower than lexical recall (98.8%), indicating that 54.2% of BM25's lexical "hits" are vocabulary-matching sentences that do not semantically entail the gold answer under NLI. Despite this, the main finding of Appendix G is strengthened, not weakened: BM25 achieves **0.0%** Stage-2 accuracy even in the 38 instructions where NLI confirms semantic entailment of the gold answer—a stronger oracle control than the 2.6% accuracy from the lexical analysis. BM25's 2 correct answers (2.4% overall) occur exclusively in the lexical-only group, where the NLI does not confirm full semantic entailment.

QCCS achieves 60.0% Stage-2 accuracy in the 10 instructions where NLI confirms semantic entailment, and 20.5% in the 73 instructions where it does not—confirming that QCCS succeeds primarily through query-conditional context alignment rather than gold-sentence retrieval in either the lexical or semantic sense.

A caveat: the NLI threshold (bidirectional entailment > 0.5 with deberta-v3-small) may be conservative for short clinical evidence strings paired with longer EHR sentences, explaining the large lex-only gap (54.2% for BM25). This calibration uncertainty does not affect the key finding, since Stage-2 accuracy is 0.0% for BM25 regardless of whether the semantic threshold is met.

L BM25 Stage-2 Accuracy at Reduced k

Motivation. The reviewer asks whether reducing k to improve context precision narrows the end-to-end accuracy gap between BM25 and QCCS. At k=20, BM25 retains 14.9 gold-vocabulary sentences on average (precision 59.5%) yet achieves only 2.4% Stage-2 accuracy. If distractors in the BM25 context suppress the reader, precision-optimized lower-k contexts may improve accuracy.

Method. BM25 is evaluated at $k \in \{1, 3, 5\}$ using the same Qwen2.5-7B-Instruct inference pipeline as the main experiment, with LLM-as-judge evaluation. Implementation: `modal run modal_app.py::run_bm25_k_sweep`. Gold-answer retrieval recall at lower k is reported in Table 4 (Stage 1 recall); Stage-2 accuracy at each k is in Table 14 below.

Table 14: BM25 Stage-2 LLM-as-judge accuracy (%) at $k \in \{1, 3, 5, 20\}$. k=20 is the main-paper result. At k=1, BM25 retrieves a single top-ranked sentence; Stage-2 accuracy tests whether distractor reduction improves downstream reasoning.

k	Overall acc (%)	Middle acc (%)
1	12.0	10.0
3	8.4	10.0
5	8.4	16.7
20	2.4	3.3
QCCS (k=20)	25.3	16.7

Interpretation. Distractor reduction provides a partial benefit: BM25 accuracy rises from 2.4% at k=20 to 12.0% at k=1 (5× improvement), indicating that fewer distractors do help the reader. However, QCCS (k=20) still achieves 25.3% overall—2.1× BM25-k=1—despite having a larger context. This establishes that the QCCS advantage is *not* solely attributable to context compactness:

query-conditional sentence selection provides benefit beyond what distractor reduction alone can explain. The middle-position gap is particularly striking: BM25-k=1 achieves 10.0% vs. QCCS 16.7% in the 30–70% position band, where CLitM suppression is strongest. Combined with the conditional accuracy finding (BM25 at k=20 achieves only 2.6% even in 78 instructions where the gold sentence is present), this confirms that BM25 retrieval quality, not context length alone, limits Stage-2 accuracy.